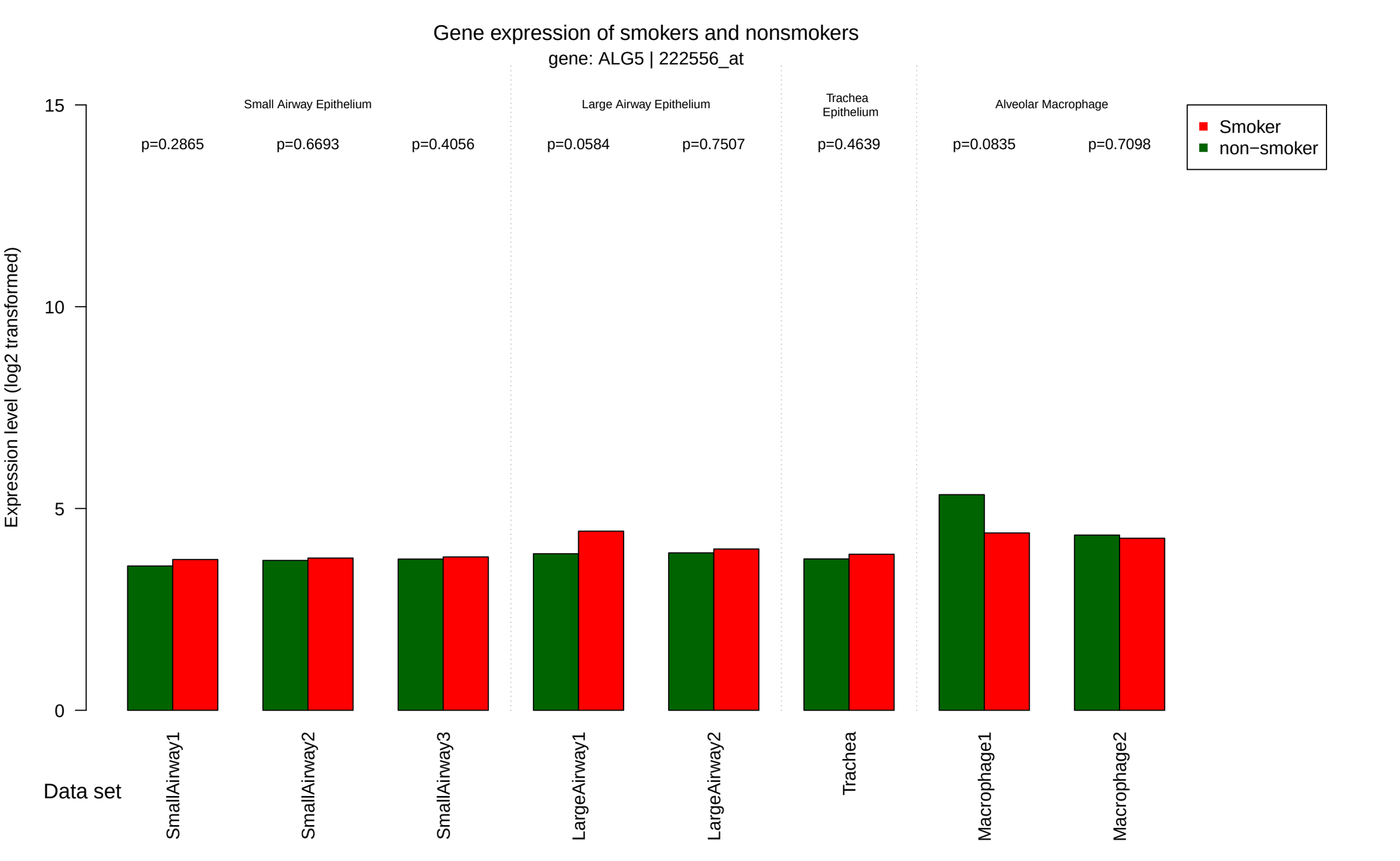
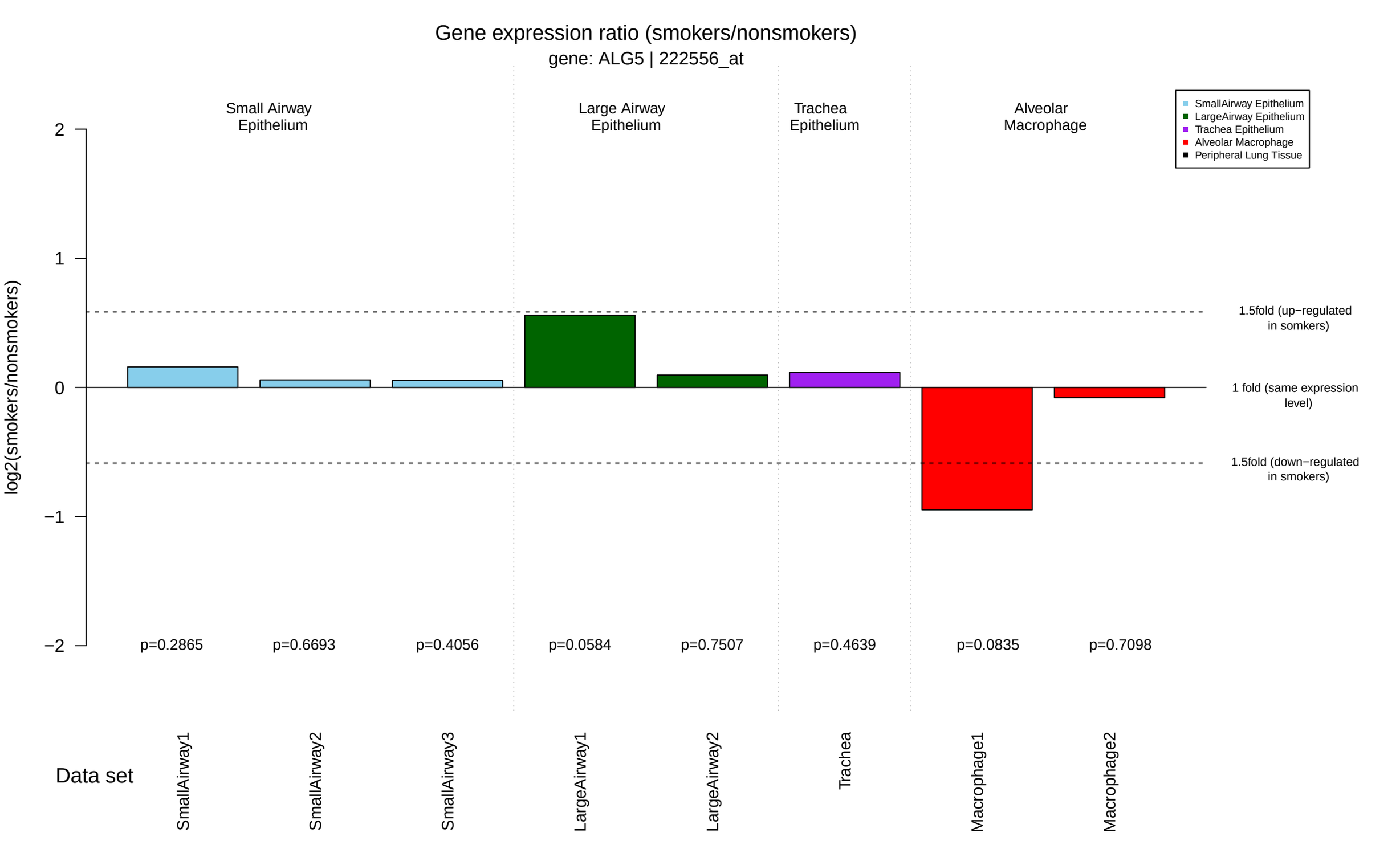
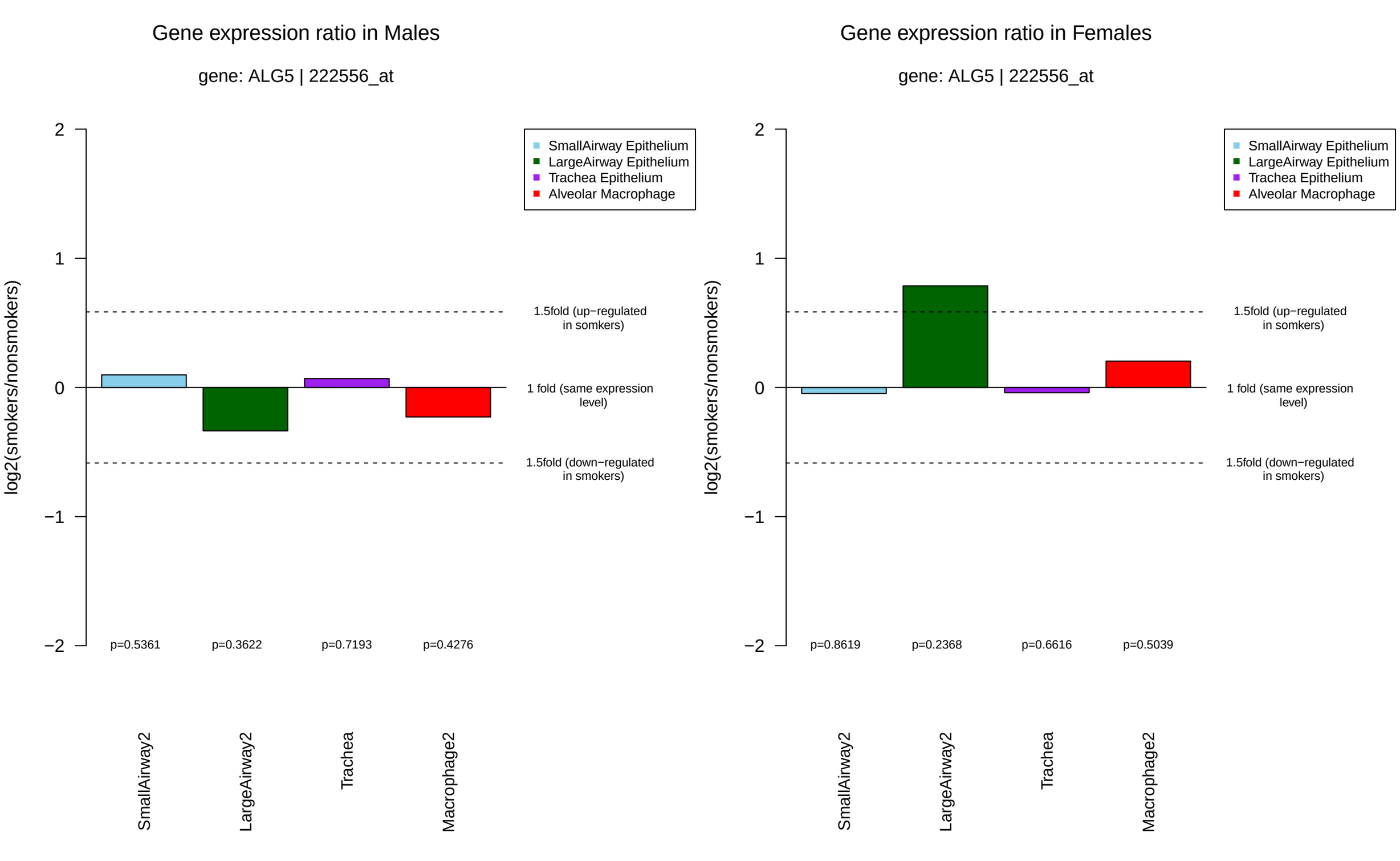
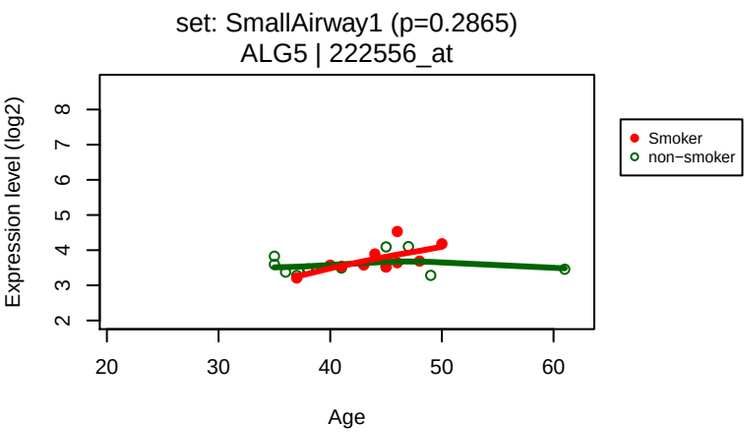
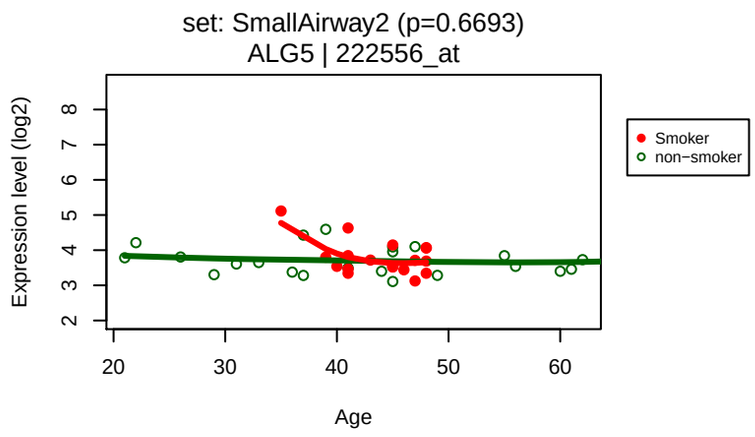
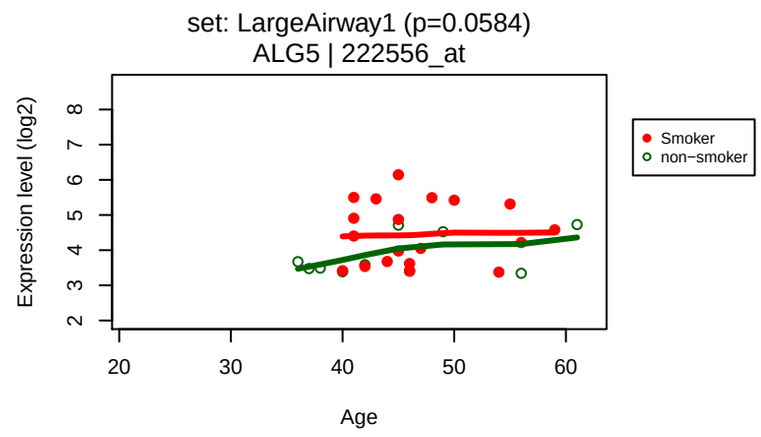

Gene expression of smokers and nonsmokers
gene: ALG5 | 222556_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2865
p=0.6693
p=0.4056
p=0.0584
p=0.7507
p=0.4639
p=0.0835
p=0.7098
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

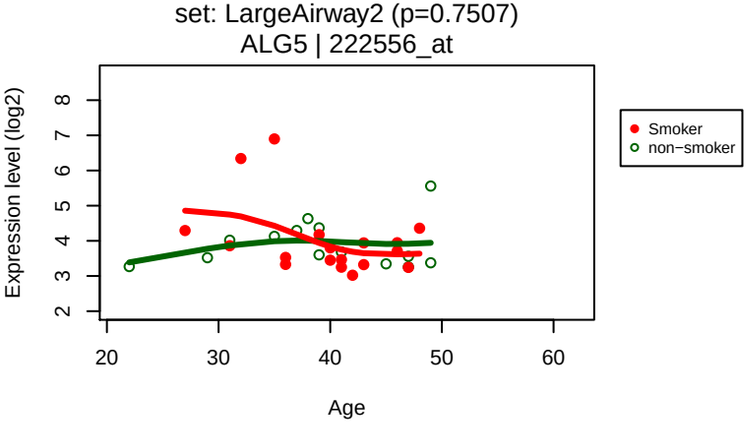

Gene expression ratio (smokers/nonsmokers)
gene: ALG5 | 222556_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2865
p=0.6693
p=0.4056
p=0.0584
p=0.7507
p=0.4639
p=0.0835
p=0.7098
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

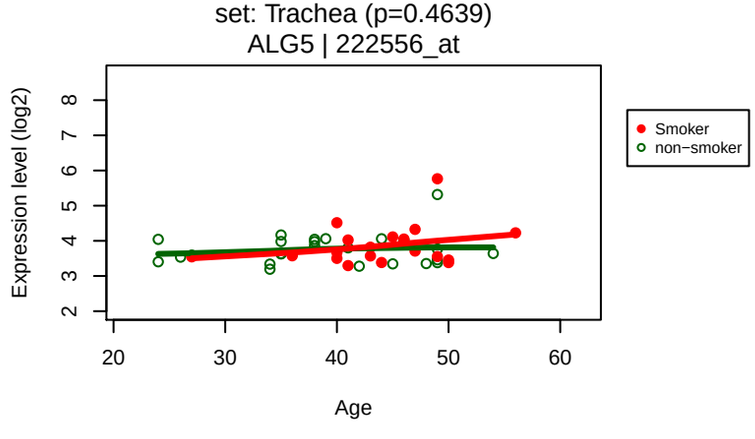

Gene expression ratio in Males
gene: ALG5 | 222556_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5361
p=0.3622
p=0.7193
p=0.4276
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALG5 | 222556_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8619
p=0.2368
p=0.6616
p=0.5039
SmallAirway2
LargeAirway2
Trachea
Macrophage2

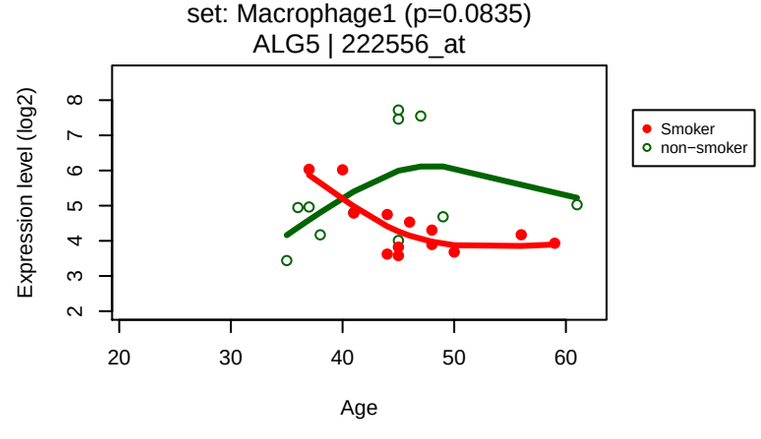

set: SmallAirway1 (p=0.2865)
ALG5 | 222556_at
Expression level (log2)
Age
Smoker
non−smoker

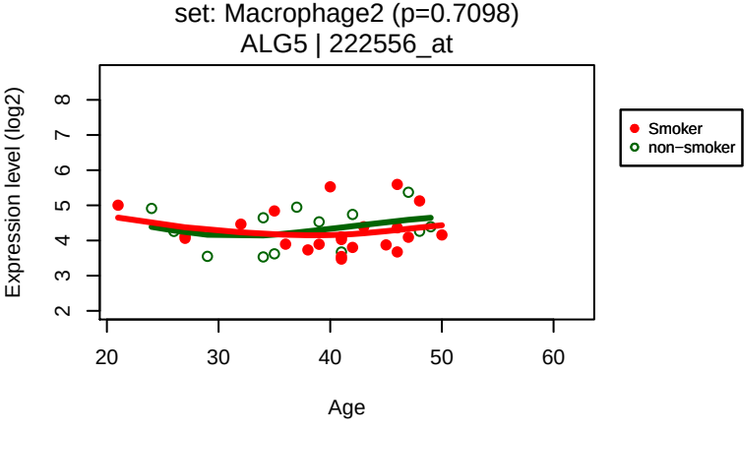

set: SmallAirway2 (p=0.6693)
ALG5 | 222556_at
Expression level (log2)
Age
Smoker
non−smoker

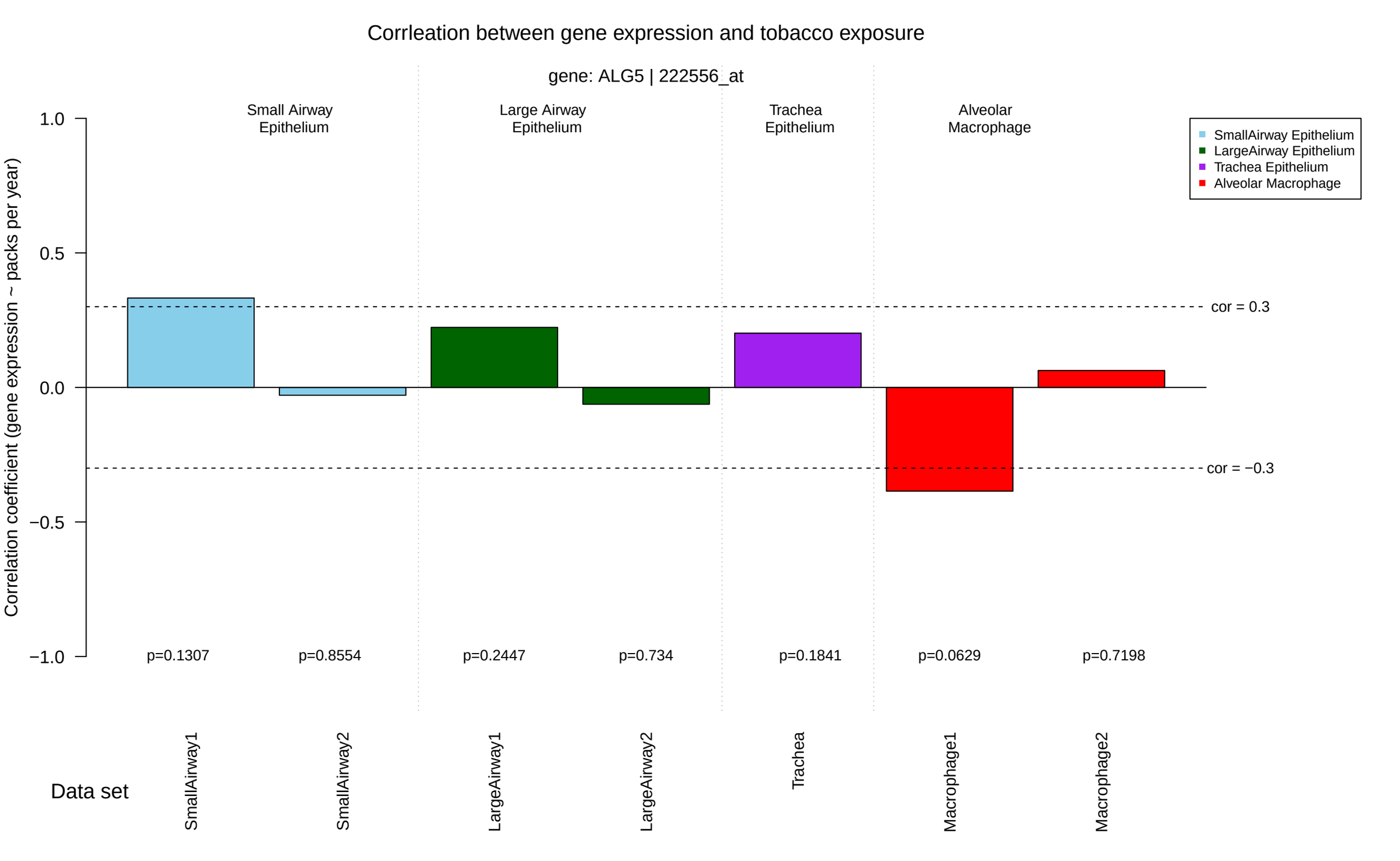

set: LargeAirway1 (p=0.0584)
ALG5 | 222556_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7507)
ALG5 | 222556_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4639)
ALG5 | 222556_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0835)
ALG5 | 222556_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7098)
ALG5 | 222556_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALG5 | 222556_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1307
p=0.8554
p=0.2447
p=0.734
p=0.1841
p=0.0629
p=0.7198
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2